# Md Shohanoor Rahman

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## Research Interests

- **Focus:** Trustworthy and explainable AI, especially testing whether explanations of image classifiers are correct.
- **Methods:** Generative models (diffusion, GANs), concept-based explanations, counterfactual evaluation.
- **Areas:** Computer vision, medical image analysis, health data.

## Technical Summary

- Research experience in testing concept explanations (TCAV, attribution) against known-target controls.
- Research experience in diffusion counterfactuals: a concept-conditioned DDPM with seven concept knobs.
- Research experience in explainable ML for clinical data with SHAP and LIME (two first-author IEEE papers).
- Research experience in GAN-based synthetic tabular data and its evaluation in eight dimensions.
- Hands-on experience in PyTorch training of CNNs, ViTs and diffusion models on multi-GPU and TPU.
- Hands-on experience in rigorous evaluation: positive and null controls, rules fixed in code before each run.
- Hands-on experience in production software engineering (3.5+ years): Spring Boot, Kafka, Kubernetes, Azure.

## Education

**Rajshahi University of Engineering & Technology (RUET)** — Rajshahi, Bangladesh
*B.Sc. in Computer Science & Engineering* — *Jan 2017 – Aug 2023*

- **Thesis:** *Generative Adversarial Network based Synthetic Data Generation System* (GitHub).
- **Relevant coursework**

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|---|---|---|
| Artificial Intelligence | Applied Statistics and Queuing Theory | Operating Systems |
| Neural Networks and Fuzzy Systems | Discrete Mathematics | Parallel and Distributed Processing |
| Data Mining | Numerical Methods | Computer Architecture and Design |
| Digital Image Processing | Data Structure | Software Engineering |
| Digital Signal Processing | Computer Algorithms | Information System Analysis and Design |
| Vector Analysis and Linear Algebra | Database Systems | Technical Writing and Presentation |

## Publications

**Preprint under journal review**

[J1] **Md. Shohanoor Rahman**, Md. Hasibul Islam, Md Kishor Morol, Dip Nandi, Liew Tze Hui. "Testing concept-level audits of melanoma classifiers with diffusion counterfactuals and matched positive controls." *arXiv:2611.02119* [cs.CV], Nov 2026. DOI: 10.48550/arXiv.2611.02119. Under review at *Scientific Reports*. Code and result tables: github.com/srahman7900/melanoma-concept-audit.

**Peer-reviewed conference papers**

[C1] **Md. Shohanoor Rahman**, Abid Hasan Mahmud, S. M. Mehedy Hasan. "Interpretable Multi-Class Obesity Classification Using Bayesian Optimization and SHAP Insights." In *Proc. 28th International Conference on Computer and Information Technology (ICCIT)*, IEEE, Cox's Bazar, Bangladesh, Dec 2025. DOI: 10.1109/ICCIT68739.2025.11491728. *Presented by the author.*

[C2] **Md. Shohanoor Rahman**, Abid Hasan Mahmud, S. M. Mehedy Hasan. "Explainable Machine Learning for Chronic Kidney Disease Prediction with SHAP and LIME." In *Proc. 28th International Conference on Computer and Information Technology (ICCIT)*, IEEE, Cox's Bazar, Bangladesh, Dec 2025. DOI: 10.1109/ICCIT68739.2025.11490146.

## Research Experience

**Testing concept-level audits of melanoma classifiers [J1]** — Dec 2025 – Present
*First author. Supervisors: Md Kishor Morol, Liew Tze Hui and Dip Nandi*

- A concept-conditioned DDPM on ISIC 2020 (128 px) in which each 7-point checklist concept is a knob from 0 to 1. Removed a signal-leak bias in the start noise (cross-validated $R^2 = 0.90$) with colour-calibrated start noise, cutting colour outliers from 78.5–84.8% to 6.6–13.3%.
- Tested the audit, TCAV and observational attribution against classifiers with known targets: 16 matched controls, 12 controls labelled by a second evaluator and 18 permuted-label nulls. Fixed each decision rule in code before its run (SHA-256 fingerprints for the later rules) and reported the first rule, which failed because its quartile controls were confounded.

- The audit found the target in 16 of 16 matched controls and in 12 of 12 second-evaluator controls (three concepts); TCAV found 11 of 16 (own implementation) and 0 of 3 (Captum 0.9.0). The pigmentation sweep raised the melanoma logit in 15 of 15 ISIC 2020 (+3.12) and 6 of 6 HAM10000 classifiers (+4.01), about four times the mean null response.

**Explainable machine learning for clinical data [C1, C2]** Bangladesh
*Group lead. Advisor: S. M. Mahedy Hasan, Assistant Professor, CSE, RUET* *Jan 2025 – Dec 2025*

- Built SHAP and LIME pipelines for chronic kidney disease prediction (4,921 patient records) and benchmarked eleven classifiers for six-level obesity classification on a new dataset of 650 people collected in Bangladesh.

**Undergraduate thesis: GAN-based synthetic tabular data** RUET, Rajshahi
*Supervisor: Barshon Sen, Assistant Professor, CSE, RUET* *Nov 2022*

- Trained CTGAN on a skewed, imbalanced financial dataset (6,819 firms, 95 ratios, 3.2% bankrupt) and evaluated the synthetic data in eight dimensions, from distributions and PCA to detection tests and train-on-synthetic, test-on-real.

## Professional Experience

**Brain Station 23 PLC** Dhaka, Bangladesh
*Software Engineer* *Mar 2023 – Present*

- **IRIS platform, Tennant Company (USA):** Azure telemetry for autonomous cleaning machines; worked on the Overview dashboard, Tech Hub (semantic search, Azure AI Search) and customer onboarding.
- **AB Direct (AB Bank) and NCC Always (NCC Bank):** As a core team member, developed and designed the architecture for two live internet banking platforms (Notable features: Fund Transfers, Bill Payment, TopUp, Service Requests).
- **Pocket:** Developed the microservice architecture for admin, customer and merchant portals of a digital wallet.

## Technical Skills

- **Machine learning:** PyTorch, TensorFlow, scikit-learn, XGBoost, CatBoost; diffusion models (DDPM, DDIM, classifier-free guidance), GANs, concept bottleneck models, CNNs and ViTs.
- **Explainability and statistics:** TCAV (Captum), SHAP, LIME; positive and null controls; Wilcoxon, Benjamini-Hochberg, bootstrap.
- **Programming and tools:** Python, Java, TypeScript, C/C++, SQL; Git, LaTeX, Kaggle (multi-GPU, TPU).
- **Software engineering:** Spring Boot microservices, Angular, Kafka, PostgreSQL, Redis, Kubernetes, Azure.
- **Languages:** Bengali (native), English (professional working proficiency).

## Training and Activities

- **DeepLearning.AI (Coursera):** GANs Specialization; four Deep Learning Specialization courses.
- **Projects:** pix2pix (aerial image to map), CycleGAN (image translation), U-Net (biomedical segmentation).
- **RUET Analytical Programming Lab:** Actively participated in programming contests, organised contests.

## Test Scores

- **GRE General:** 312 (Quantitative 165, Verbal 147, Analytical Writing 3.0) Jul 2025

## References

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